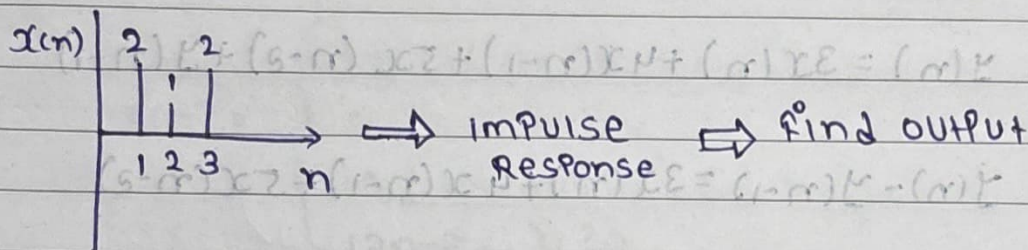


Question-1



input signal: $x(n] = 2\delta(n-1) + 1\delta(n-2) + 2\delta(n-3)$

Impulse Response: $h(n] = u(n] - u(n-5)$

$$h(n] = 1 \quad 0 \leq n \leq 4$$

$$y(n] = x(n] * h(n]$$

$$y(n] = \sum_{k=-\infty}^{\infty} x(k] h(n-k]$$

$$x(n] = 1, 2, 3$$

$$y(n] = 2h(n-1) + 1h(n-2) + 2h(n-3)$$

$$h(n-1] = 1 \text{ for } 1 \leq n \leq 5$$

$$h(n-2] = 1 \text{ for } 2 \leq n \leq 6$$

$$h(n-3] = 1 \text{ for } 3 \leq n \leq 7$$

$$n=1 \quad y(1] = 2$$

$$n=2 \quad y(2] = 2+1=3$$

$$n=3 \quad y(3] = 2+1+2=5$$

$$n=4 \quad y(4] = 2+1+2=5$$

$$n=5 \quad y(5] = 2+1+2=5$$

$$n=6 \quad y(6] = 1+2=3$$

$$y(n) = \{0, 2, 3, 5, 5, 3, 2, 0\}$$

Question -2)

$$y(n) = 3x(n) + 4x(n-1) + 5x(n-2) + y(n-1)$$

$$y(n) - y(n-1) = 3x(n) + 4x(n-1) + 5x(n-2)$$

a) impulse response $h(n)$

$$H(z) = \frac{3 + 4z^{-1} + 5z^{-2}}{1 - z^{-1}}$$

$$\frac{1}{1 - z^{-1}} \xleftrightarrow{Z^{-1}} y(n)$$

$$h(0) = 3, h(1) = 3 + 4 = 7, h(2) = 3 + 4 + 5 = 12$$

$$h(n) = \begin{cases} 0, & n < 0 \\ 3, & n = 0 \\ 7, & n = 1 \\ 12, & n \geq 2 \end{cases}$$

b) unit step response $s(n)$

$$S(z) = H(z)X(z)$$

$$S(z) = \frac{3 + 4z^{-1} + 5z^{-2}}{(1 - z^{-1})^2}$$

$$n = 0 : s(0) = 3$$

$$n = 1 : s(1) = 3(2) + 4(1) = 10$$

$$s(n) = 3(n+1) + 4n + 5(n-1)$$

$$s(n) = 12n - 2 \quad (n \geq 2)$$

$$s(n) = \begin{cases} 0, & n < 0, \\ 3, & n = 0, \\ 10, & n = 1, \\ 12n - 2, & n \geq 2 \end{cases}$$

c) System Function $H(z)$

$$y(z) - z^{-1}y(z) = (3 + 4z^{-1} + 5z^{-2})x(z)$$

$$H(z) = \frac{y(z)}{x(z)} = \frac{3 + 4z^{-1} + 5z^{-2}}{1 - z^{-1}}$$

$$H(z) = \frac{3z^2 + 4z + 5}{z^2 - z} = \frac{3z^2 + 4z + 5}{z(z-1)}$$

d) Frequency Response $H(e^{j\omega})$

$$H(e^{j\omega}) = \frac{3 + 4e^{-j\omega} + 5e^{-j2\omega}}{1 - e^{-j\omega}}$$

e) Magnitude Response $|H(e^{j\omega})|$

$$|H(e^{j\omega})| = \frac{|3 + 4e^{-j\omega} + 5e^{-j2\omega}|}{|1 - e^{-j\omega}|}$$

$$= \frac{\sqrt{(3 + 4\cos\omega + 5\cos 2\omega)^2 + (4\sin\omega + 5\sin 2\omega)^2}}{2|\sin(\frac{\omega}{2})|}$$

f) Phase Response $\angle H(e^{j\omega})$

$$\angle H(e^{j\omega}) = \angle (3 + 4e^{j\omega} + 5e^{-j2\omega}) - \angle (1 - e^{-j\omega})$$

$$\angle (1 - e^{-j\omega}) = -\omega/2 + \pi/2$$

$$\tan^{-1} \left(\frac{3\sin\omega + 5\sin 2\omega}{3 + 4\cos\omega + 5\cos 2\omega} \right)$$

$$\angle H(e^{j\omega}) = \tan^{-1} \left(\frac{4\sin\omega + 5\sin 2\omega}{3 + 4\cos\omega + 5\cos 2\omega} \right) - \left(-\frac{\omega}{2} + \frac{\pi}{2} \right)$$

g) $x(n) = e^{3n} y(n)$

$$Y(z) = \frac{3z^2 + 4z + 5}{(z-1)(z-8)}$$

$$Y(z) = 3 + \frac{A}{z-1} + \frac{B}{z-8}$$

$$A = \frac{-12}{8-1}, \quad B = 7 + 3 \cdot 8 + \frac{12}{8-1}$$

$$y(n-1) = \frac{1}{z-1} \quad 8^{n-1}(n-1) = \frac{1}{z-8}$$

$$y(n) = 3\delta(n) + Ay(n-1) + B8^{n-1}y(n-1)$$

$$H(z) = \frac{4z^{-1} + 5z^{-2}}{1 - z^{-1}}$$

Question-3)

$$F(x, y) = \begin{bmatrix} 1 & 5 & 7 \\ 2 & 3 & 9 \\ 4 & 6 & 2 \end{bmatrix} \quad (3 \times 3) \quad L=8$$

a) $F(1, 2) = 5$

b) $F(3, 2) = 6$

c) Grey level resolution

8 levels = 2, 1, 2, 3, 4, 5, 6, 7, 93

$$\log_2(8) = 3 \text{ bits}$$

d) size of image

$$3 \times 3 = 9 \text{ pixels}$$

e) Zoom by Factor 2 - Pixel Replication

$$(3 \cdot 2) \times (3 \cdot 2) = 6 \times 6$$

$$\begin{bmatrix} 1 & 1 & 5 & 5 & 7 & 7 \\ 1 & 1 & 5 & 5 & 7 & 7 \\ 2 & 2 & 3 & 3 & 9 & 9 \\ 2 & 2 & 3 & 3 & 9 & 9 \\ 4 & 4 & 6 & 6 & 2 & 2 \\ 4 & 4 & 6 & 6 & 2 & 2 \end{bmatrix}$$

g) zoom by factor 2-k times zooming

$$k=2$$

$$(6 \times 6)$$

2x zoom

$$\begin{bmatrix} 1 & 3 & 5 & 6 & 7 & 7 \\ 1 & 2.75 & 4 & 6 & 8 & 8 \\ 2 & 2.50 & 3 & 6 & 9 & 9 \\ 3 & 2.75 & 4.50 & 5 & 5 & 5 \\ 4 & 5 & 6 & 4 & 2 & 2 \\ 4 & 5 & 6 & 4 & 2 & 2 \end{bmatrix}$$

Question-4)

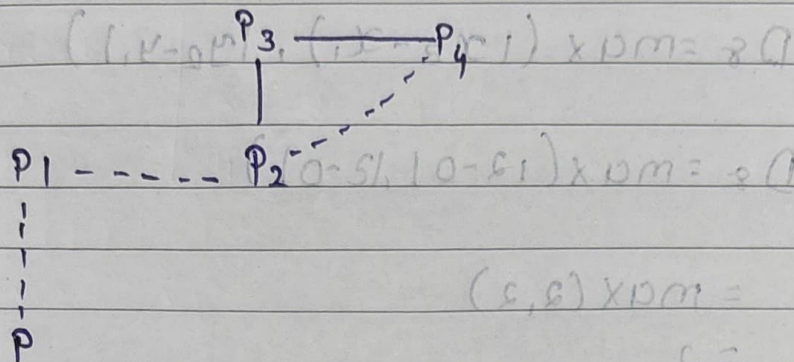
$$F(x,y) = \begin{bmatrix} 1 & 8 & 9 & 7 & 9 \\ 2 & 3 & 4 & 6 & 7 \\ 4 & 3 & 2 & 6 & 7 \\ 8 & 5 & 6 & 7 & 8 \\ 9 & 6 & 2 & 5 & 4 \end{bmatrix}$$

a) order of image = 5×5

b) shrink the image

$$\lceil 5/2 \rceil \times \lceil 5/2 \rceil = 3 \times 3$$

$$\begin{bmatrix} 1 & 9 & 9 \\ 4 & 2 & 7 \\ 9 & 2 & 4 \end{bmatrix}$$

Question-6)4) perfect sampling

$$P_1 = (0, 0)$$

$$P_2 = (2, 0)$$

$$P_3 = (2, 2)$$

$$P_4 = (4, 2)$$

$$P \rightarrow P_1 \quad (0, -2)$$

$$P(1) = (0, 0)$$

$$P_3(2, 2)$$

a) Euclidean distance

$$D_E = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$D_E = \sqrt{(2-0)^2 + (2-0)^2}$$

$$= \sqrt{4+4}$$

$$= \sqrt{8}$$

$$= 2.828$$

b) city block distance

$$D_4 = |x_2 - x_1| + |y_2 - y_1|$$

$$D_4 = |2-0| + |2-0| = 2+2 = 4$$

c) Chessboard distance

$$D_s = \max(|x_2 - x_1|, |y_2 - y_1|)$$

$$D_s = \max(|2 - 0|, |2 - 0|) \dots \dots \dots 2$$

$$= \max(2, 2)$$

$$= 2$$

Question 6)

$$0 \quad 1 \quad 1$$

$$0 \quad 1 \quad 0$$

$$0 \quad 0 \quad 1$$

$$(0, 0) = 1$$

$$(0, 1) = 1$$

$$(1, 1) = 1$$

$$(1, 2) = 1$$

$$(2, 0) = 1$$

$$P = 1$$

$$N_4(P) \rightarrow K P$$

$$N_D(P) \rightarrow \text{Diagonal}$$

$$N_8(P) \rightarrow 8 \cdot P(N_4 + N_D)$$

$$1) N_4(P)$$

$$(1, 2) \rightarrow 1$$

$$(3, 2) \rightarrow 0 \quad (0-2) + (0-2) \sqrt{2} = 0$$

$$(2, 1) \rightarrow 0$$

$$(2, 3) \rightarrow 0$$

$$N_4(P) = 2(1, 2)$$

$$2) N_D(P)$$

$$(1, 1) = 0 \quad (1-3) + (1-1) \sqrt{2} = 0$$

$$(1, 3) = 1 \quad (3, 3) = 1$$

$$N_D = 2 + 0 = (0-2) + (0-2) \sqrt{2} = 0$$

$$N_0(p) = \{(1,3), (3,3)\}$$

$$3) N_8(p)$$

$$N_8(p) = \{(1,2), (1,3), (3,3)\}$$